

Creating the Art-Directed Groom for Legend in Disney's "Strange World"

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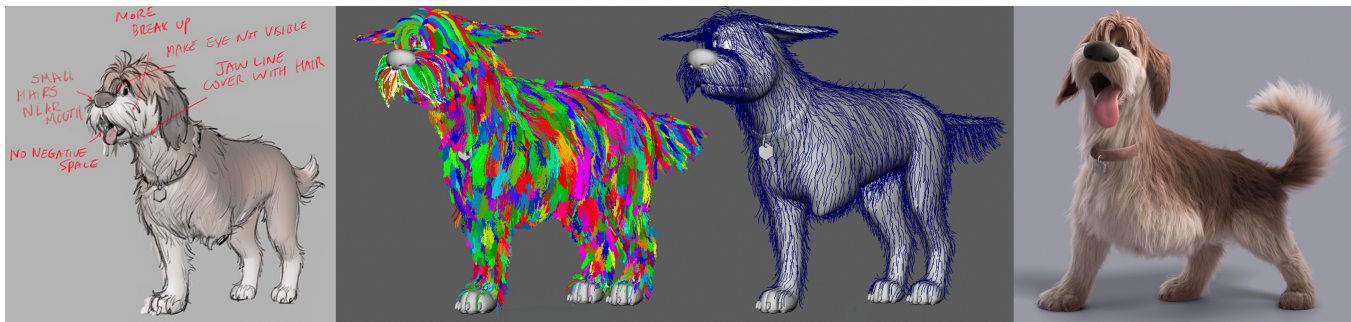


Figure 1: Example image of Legend's groom notes, final groom regions, final guides, and render. ©2023 Disney

ABSTRACT

Translating the design language of Walt Disney Animation Studios' "Strange World" onto a fur groom for the Clade family's lovable dog, Legend, at a level of scrutiny that is often reserved for our human characters led us to exploration outside of our regular workflows. The groom had to adhere to the artistic vision of the Visual Development and Animation teams while still working within our pipeline and staying consumable by downstream departments.

We were able to leverage Houdini to supplement Disney's XGen to meet the aesthetic needs of the show. With Houdini we were able to quickly put down fur similar to our map-based grooming methods and because we folded that into a similar workflow as our in-house hierarchical grooming tool, Tonic, we were able to maintain the ability to fine-tune at a more granular level.

Through thoughtful collaboration we were able to deliver a fun and unique groom while exploring new techniques that will open up new possibilities for grooming workflows moving forward.

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1 INTRODUCTION

Legend's inspiration was drawn from a mixture of large, shaggy sheep-dogs with quirky personalities. The coats of these types of dogs tend to have a considerable amount of messiness, noise, and personality in the individual clumps of hair. In order to stylize this natural phenomenon and adhere to the French comic style, the chaos and individuality of these clumps would have to be amplified or tamped down in specific areas. We also wanted to make Legend look as much as possible like Jaegar, the estranged patriarch of the family, in order to set up a comical dynamic.

In order to increase the cuteness, Legend had to be able to act and emote right alongside his human counterparts. This led to an increased focus by animators and art directors on the visual clarity of his major emotional features: his eyes, mouth, tail and ears. Legend's groom would also need sufficient fur coverage to account for any squash and stretch that would be introduced by the animation style of the movie.

Our in-house grooming tool set includes *iGroom*, a brush-based workflow in Disney's XGen that produces maps controlling procedural fur generation, and *Tonic* [Kaur et al. 2018], a hierarchical tube-based sculpting tool that provides precise control over the shape of individual locks of hair. Tonic is usually the tool of choice for artists when a high level of creative control is required. Tonic's interactive sculpting workflow allows artists to address the characteristics and shapes of individual locks of hair, e.g. within stylized bangs or ponytails.

Creating tubes for every clump of hair on Legend's full groom would be too time-consuming. While the *iGroom* workflow is efficient and is well-suited for characters with short fur, the *iGroom* tool and associated output maps do not offer the level of control necessary to achieve the mixed-length fur design of Legend. To achieve the furry scale and creative specificity of Legend's groom, we needed an approach that would allow us to quickly lay down

groom guide curves en masse but that would also let us zoom in on specific areas and craft the shape of individual clumps of fur.

2 HYBRID GROOMING APPROACH

The general philosophy of the Look Department is to provide artists with the most versatile ways to achieve our grooms, and so we are constantly analyzing the tools available to us. For the challenges presented by Legend's design, we adopted a hybrid approach leveraging Houdini's proceduralism, integrated into our grooming pipeline built around Disney's XGen.

In our standard guide-based hair pipeline, guide curves and region maps are output from Tonic, and used as input to Disney's XGen where procedurally generated hair primitives are interpolated from the guide curves, and the region maps are used to control clumping and motion. For Legend, Houdini instead of Tonic was used to generate the guide curves and region maps. By matching the output of Tonic, it was possible to leverage the features of Disney's XGen and seamlessly integrate back into the rest of our standard pipeline.

Using Houdini and the GroomBear plug-in, we were able to quickly lay down adjustable, medium-length fur guide curves. For the region maps, we created a custom Houdini node to generate the maps defining the region of influence of each guide using geodesic distance. Using geodesic, rather than Euclidean, distance is crucial especially for thin and close areas such as the ears and the lip. The guides and region maps are exported from Houdini and fed into hair descriptions in Disney's XGen. For shorter fur, iGroom was used in the areas around the face, mouth, and eyes.

Using Disney's XGen to generate the final fur primitives provided the groom artists with a familiar workflow to fine-tune the individual characteristics of the fur groom, and allowed them to work with the art director at both the clump and individual hair level. To address shape notes, the artist could jump back into Houdini and adjust specific guides.

The mixed-length style of Legend's groom combined with the expressive, energetic animation style required simulation of the medium to long hair over his whole body. This imposed additional requirements on the groom which were addressed with tight collaboration with the Simulation team. Look artists needed to finesse areas like the ears and tail to avoid interpolation issues when the regions wrapped around high curvature areas and did not have enough guides to interpolate properly. The groom artist delivered overall higher guide density guides in these areas to provide the Simulation team with sufficient curve density (Fig 2).

In addition, an iGroom description was added to areas of high squash and stretch to define an *undercoat* that helped hide any gaps that might show up in motion. One constant workflow challenge that further complicated the process was the fact that we were grooming Legend in a neutral pose which required a simulation pass to drop Legend's ears down with gravity before we could assess any grooming issues in these areas (Fig 3).

3 CONCLUSION

Using Houdini in combination with Disney's XGen allowed artists to explore new avenues for fur grooming while providing a familiar space for artists to interact and iterate with the art directors. This

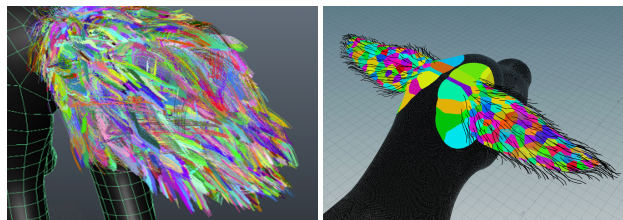


Figure 2: Left: Interpolation confusion in areas of large curvature. Right: Houdini region and guides showing adjusted guide density. ©2023 Disney



Figure 3: Simulation test of ear drooping to evaluate groom. ©2023 Disney



Figure 4: Legend walking upright with final groom. ©2023 Disney

new workflow allowed artists to quickly iterate on complex fur grooms and assess the validity of their data with downstream departments. By opening the door on this new workflow, Houdini can now be further implemented into our grooming process which will allow us to leverage the flexibility of Houdini while supplementing the artistic directability of Disney's XGen (Fig 4).

REFERENCES

- Avneet Kaur, Maryann Simmons, and Brian Whited. 2018. Hierarchical Controls for Art-Directed Hair at Disney. In *ACM SIGGRAPH 2018 Talks* (Vancouver, British Columbia, Canada) (SIGGRAPH '18). Association for Computing Machinery, New York, NY, USA, Article 13, 2 pages. <https://doi.org/10.1145/3214745.3214774>